



Chemical Theory II

Statistical

Thermodynamics

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The course is developed based on the
Cambridge chemistry courses MELT

Summary of last lecture

- Transition state theory
 - Quasi-equilibrium between reactants and transition state: K^*
 - First-order decay of transition state to products
 - Factorise reactive mode out of q_{vib}
 - Estimating steric factors by counting contributions of each type to q
- Kinetic isotope effect
 - ν goes as $\mu^{-1/2}$ so q_{vib} goes up and $k_{2\text{nd}}$ goes down for heavier isotopes.

Thermodynamic formulation of transition state theory

- As we have seen, *the problem with transition state theory is that although it offers the possibility of calculating rate constants ab initio, in practice such calculations are rather difficult.*
- An alternative approach is to utilise the basic ideas of the theory but express them in the language of classical thermodynamics. This leads to a formulation of the problem which is not predictive but is very useful for thinking about the factors which influence rate constants, especially for reactions in solution.

Thermodynamic formulation of transition state theory

- We start from the equation:

$$k_{2\text{nd}} = \frac{1}{c^\circ} k_{1\text{st}} K^*$$

- And, K^* can be rewritten

$$K^* = q^\ddagger K^\ddagger$$

- To a good approximation, the partition function for vibration along the reaction coordinate, $q^\ddagger$, is given by

$$q^\ddagger = \frac{kT}{h\nu^\ddagger}$$

- Putting these all together, the expression for $k_{2\text{nd}}$ becomes

$$k_{2\text{nd}} = \frac{1}{c^\circ} k_{1\text{st}} \frac{kT}{h\nu^\ddagger} K^\ddagger$$

- As before, we conveniently identify $k_{1\text{st}}$ with $\nu^\ddagger$ to give what is known as the Eyring equation

$$k_{2\text{nd}} = \left(\frac{1}{c^\circ} \right) \frac{kT}{h} K^\ddagger.$$

Thermodynamic formulation of transition state theory

- The next step is to use classical thermodynamics to write the equilibrium constant in terms of $\Delta_r G^\circ$:

$$\Delta_r G^{\circ,\ddagger} = -RT \ln K^\ddagger$$

where $\Delta_r G^{\circ,\ddagger}$ is the change in Gibbs energy, under standard conditions, when the transition state is formed from the reactants.

- *We have to be very careful in using this expression as it assumes that the same standard state is used in both the definition of $K^\ddagger$ and of $\Delta_r G^{\circ,\ddagger}$.* Normally, the standard state chosen for $\Delta_r G^\circ$ is, for gases, a pressure of 1 bar and, for solutions, a concentration of 1 mol dm⁻³; however, we have so far defined $K^\ddagger$ using a standard state of 1 molecule m⁻³, which is not the usual standard state for $\Delta_r G^\circ$.
- Sorting out this difficulty is just a question of a change of standard states.

Thermodynamic formulation of transition state theory

- So for a gas phase reaction, is

$$\text{gas-phase } k_{2\text{nd}} / (\text{N m}^{-2})^{-1} \text{ s}^{-1} = \left(\frac{1}{p^\circ} \right) \frac{kT}{h} \exp \left(\frac{-\Delta_r G^{\circ,\ddagger}}{RT} \right)$$

where p° is the standard pressure of 1 bar or 10^5 N m^{-2} , and $\Delta_r G^{\circ,\ddagger}$ is referred to this standard state. *Note the units of $k_{2\text{nd}}$.*

- For a reaction in solution the result has a similar form

$$\text{solution } k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} = \left(\frac{1}{[\]^\circ} \right) \frac{kT}{h} \exp \left(\frac{-\Delta_r G^{\circ,\ddagger}}{RT} \right)$$

where $[\]^\circ$ is the standard concentration of 1 mol dm^{-3} , and $\Delta_r G^{\circ,\ddagger}$ is referred to this standard state. Once again, note the units of $k_{2\text{nd}}$.

Thermodynamic formulation of transition state theory

- The standard Gibbs energy change can be written in terms of a standard enthalpy change and a standard entropy change:

$$\Delta_r G^{\circ, \ddagger} = \Delta_r H^{\circ, \ddagger} - T \Delta_r S^{\circ, \ddagger}$$

- Using this, the expression for the rate constant in solution becomes

$$k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} = \frac{1}{[\]^{\circ}} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ, \ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ, \ddagger}}{RT}\right)$$

- *This is the equation which we use to interpret rate constants in this thermodynamic formulation.*

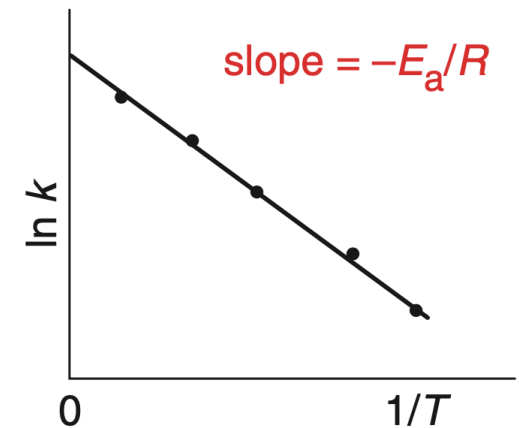
Relationship to the Arrhenius equation

$$k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} = \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ,\ddagger}}{RT}\right)$$

- This equation is very reminiscent of the Arrhenius equation:

$$k_{2\text{nd}} = A \exp\left(\frac{-E_a}{RT}\right)$$

- Tempting though it is to identify E_a as $\Delta_r H^{\circ,\ddagger}$, this would be wrong because it does not take account of the formal definition of the activation energy, which is as follows. *The slope (at a particular temperature) of a plot of $\ln k_{2\text{nd}}$ against $1/T$ has the value $-E_a/R$; this defines the activation energy.* In addition, the rate constants must be recorded at constant volume.



The activation energy is defined from the slope of a plot of $\ln k$ against $1/T$; the slope may vary as a function of temperature.

Relationship to the Arrhenius equation

- In the language of calculus this definition becomes

$$\left(\frac{\partial \ln k_{2\text{nd}}}{\partial (1/T)} \right)_V = -\frac{E_a}{R}$$

or, entirely equivalently,

$$\boxed{\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{E_a}{RT^2}.$$

- The simplest way to find E_a in terms of the other parameters used in transition state theory is to differentiate the equation of $k_{2\text{nd}}$

$$\boxed{k_{2\text{nd}} = \left(\frac{1}{c^\circ} \right) \frac{kT}{h} K^\ddagger.$$

first having taken logarithms of both sides

$$k_{2\text{nd}} = \left(\frac{1}{c^\circ} \right) \frac{kT}{h} K^\ddagger$$

$$\ln k_{2\text{nd}} = \ln \frac{k}{c^\circ h} + \ln T + \ln K^\ddagger$$

$$\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{1}{T} + \left(\frac{\partial \ln K^\ddagger}{\partial T} \right)_V$$

Reactions in solution

- The variation of the equilibrium constant with temperature is given by the familiar van't Hoff isochore

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\circ}{RT^2}$$

- Reactions in solution are little affected by changes in pressure, so we can safely use this relationship to substitute for $(\partial \ln K^\ddagger / \partial T)_V$ to give

$$\left(\frac{\partial \ln k_{2nd}}{\partial T} \right)_V = \frac{1}{T} + \left(\frac{\partial \ln K^\ddagger}{\partial T} \right)_V = \frac{1}{T} + \frac{\Delta_r H^{\circ, \ddagger}}{RT^2}$$

- Comparing this to the equation

$$\boxed{\left(\frac{\partial \ln k_{2nd}}{\partial T} \right)_V = \frac{E_a}{RT^2}}$$

we have

$$\frac{E_a}{RT^2} = \frac{1}{T} + \frac{\Delta_r H^{\circ, \ddagger}}{RT^2}$$

$$\text{hence } E_a = \Delta_r H^{\circ, \ddagger} + RT \quad (\text{in solution})$$

Gas phase reactions

- For gas phase reactions the situation is a little more complex as the van't Hoff isochore applies only to the equilibrium constant expressed in terms of partial pressures, K_p :

$$\frac{d \ln K_p}{dT} = \frac{\Delta_r H^\circ}{RT^2}$$

- Recall that $K^\ddagger$ is expressed in terms of concentrations, specifically in *molecules per unit volume*. The relationship between K_p and the equilibrium constant in terms of concentrations, K_c , is easily worked out for a particular reaction:



$$K_p = \frac{p_C p^\circ}{p_A p_B}$$

$$\text{but } p_i = \frac{n_i RT}{V} = \frac{N_i kT}{V} = c_i kT$$

$$\text{so } K_p = \left(\frac{c_C}{c_A c_B} \right) \frac{p^\circ}{c^\circ kT} = K_c \frac{p^\circ}{c^\circ kT}$$

Gas phase reactions

$$K_p = \left(\frac{c_C}{c_A c_B} \right) \frac{p^\circ}{c^\circ k T} = K_c \frac{p^\circ}{c^\circ k T}$$

- where for species i p_i is the partial pressure, n_i is the number of moles, N_i is the number of molecules and c_i is the concentration in molecules per unit volume; we have also used $R = N_A k$. So, substituting this expression for K_p into the van't Hoff isochore we have

$$\begin{aligned} \frac{d}{dT} \left(\ln K_c - \ln T + \ln \frac{p^\circ}{k c^\circ} \right) &= \frac{\Delta_r H^\circ}{RT^2} \\ \frac{d \ln K_c}{dT} - \frac{1}{T} &= \frac{\Delta_r H^\circ}{RT^2} \\ \frac{d \ln K_c}{dT} &= \frac{\Delta_r H^\circ + RT}{RT^2} \end{aligned}$$

- Then using this we can have

$$\begin{aligned} \left(\frac{\partial \ln k_{2nd}}{\partial T} \right)_V &= \frac{1}{T} + \left(\frac{\partial \ln K^\ddagger}{\partial T} \right)_V \\ &= \frac{1}{T} + \frac{\Delta_r H^{\circ, \ddagger} + RT}{RT^2} \\ &= \frac{\Delta_r H^{\circ, \ddagger} + 2RT}{RT^2}. \end{aligned}$$

Gas phase reactions

$$\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{E_a}{RT^2}.$$

$$\left(\frac{\partial \ln k_{2\text{nd}}}{\partial T} \right)_V = \frac{\Delta_r H^{\circ, \ddagger} + 2RT}{RT^2}.$$

- Comparing the two equations we see that for the bimolecular gas phase reaction

$$E_a = \Delta_r H^{\circ, \ddagger} + 2RT \quad (\text{bimolecular, gas phase}).$$

- Admittedly, this discussion is rather convoluted and leads to no startling conclusion. It is important, however, to realise that the enthalpy of activation is not the same as the activation energy.

The pre-exponential or A-factor

- Returning to the equation

$$k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} = \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ,\ddagger}}{RT}\right)$$

and substituting in for $\Delta_r H^{\circ,\ddagger}$ we have, for a reaction in solution,

$$\begin{aligned} k_{2\text{nd}} &= \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-E_a + RT}{RT}\right) \\ &= \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) e^1 \exp\left(\frac{-E_a}{RT}\right) \\ &= \left[\frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R} + 1\right) \right] \exp\left(\frac{-E_a}{RT}\right) \end{aligned}$$

where, for convenience, we have put the factor of e into the first exponential. By comparison with the Arrhenius equation, the quantity in the square bracket can be equated with the pre-exponential or **A**-factor. *We see that the entropy of activation is the significant contributor to the A-factor.*

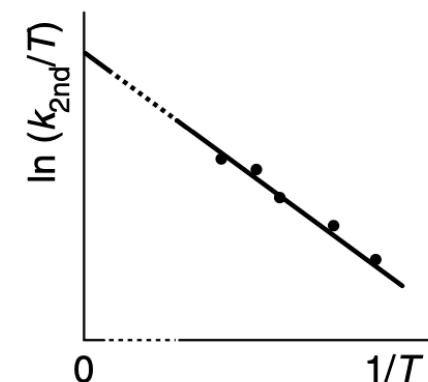
Experimental determination of the enthalpy and entropy of activation

$$k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} = \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ,\ddagger}}{RT}\right)$$

- This equation above can be rearranged to give a straight line by taking logarithms

$$\begin{aligned} k_{2\text{nd}} &= \frac{1}{[\]^\circ} \frac{kT}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ,\ddagger}}{RT}\right) \\ \frac{k_{2\text{nd}}}{T} &= \frac{1}{[\]^\circ} \frac{k}{h} \exp\left(\frac{\Delta_r S^{\circ,\ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ,\ddagger}}{RT}\right) \\ \ln \frac{k_{2\text{nd}}}{T} &= \ln \frac{k}{h[\]^\circ} + \frac{\Delta_r S^{\circ,\ddagger}}{R} - \frac{\Delta_r H^{\circ,\ddagger}}{R} \left(\frac{1}{T}\right) \end{aligned}$$

- So a plot of $\ln (k_{2\text{nd}}/T)$ against $(1/T)$ should yield a straight line of slope $-\Delta_r H^{\circ,\ddagger}/R$ and intercept $(\ln k / h[\]^\circ + \Delta_r S^{\circ,\ddagger}/R)$; this is sometimes called *an Eyring plot*. In making this plot it is implicit that $\Delta_r H^{\circ,\ddagger}$ and $\Delta_r S^{\circ,\ddagger}$ are independent of temperature. Note also that we are assuming that $[\]$ is 1 mol dm^{-3} and $k_{2\text{nd}}$ is in units of $\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$, so that *the thermodynamic quantities are referred to this same standard state of 1 mol dm^{-3} .*

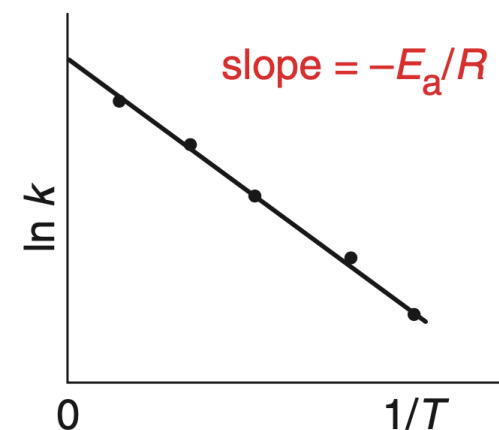


An Eyring plot used to determine the enthalpy and entropy of activation. Note that a long extrapolation is needed to find the intercept, so this is likely to be inaccurate.

Experimental determination of the enthalpy and entropy of activation

- An alternative is to make the usual Arrhenius plot, determine E_a and A , and then relate these to determine $\Delta_r H^{\circ,\ddagger}$ and $\Delta_r S^{\circ,\ddagger}$ using the relationships given in previous slides.
- Typically, measurements of the rate constant are only available for a relatively small temperature range, and so a long extrapolation is needed to find the intercept and hence $\Delta_r S^{\circ,\ddagger}$; therefore the numerical value is often subject to considerable error.

$$k_{2\text{nd}} = A \exp\left(\frac{-E_a}{RT}\right)$$



The activation energy is defined from the slope of a plot of $\ln k$ against $1/T$; the slope may vary as a function of temperature.

The effect of pressure: volumes of activation

- Altering the pressure of a gas-phase reaction has a direct effect on the rate of the reaction simply as the concentrations of the species are directly proportional to the pressure. For a reaction in solution, altering the pressure on the solvent does not affect the concentration yet it is found experimentally that at sufficiently high pressures (up to 10,000 atmospheres) reaction rates are affected. *The change in the rate is attributed to a change in the rate constant as a result of the change in pressure. The rate constant may go up or down in response to an increase in the pressure,* and it is found that the largest changes are seen for reactions involving ions in polar solvents.

The effect of pressure: volumes of activation

- The thermodynamic formulation of transition state theory can be used to interpret these effects. We start out with one of the Master Equations

$$dG = Vdp - SdT$$

and impose the condition of constant temperature, which gives us

$$dG = Vdp \quad (\text{const. temp.}) \quad \text{or} \quad \left(\frac{\partial G}{\partial p} \right)_T = V.$$

- Starting from the equation of $k_{2\text{nd}}$ for a reaction in solution and taking logarithms we have

$$k_{2\text{nd}} = \frac{1}{[\]^\circ} \frac{kT}{h} \exp \left(\frac{-\Delta_r G^{\circ, \ddagger}}{RT} \right)$$
$$\ln k_{2\text{nd}} = \ln \frac{kT}{[\]^\circ h} - \frac{\Delta_r G^{\circ, \ddagger}}{RT}.$$

The effect of pressure: volumes of activation

- To find the variation in k_{2nd} with pressure at constant temperature we form the appropriate partial derivative:

$$\left(\frac{\partial \ln k_{2nd}}{\partial p}\right)_T = \frac{-1}{RT} \left(\frac{\partial \Delta_r G^{\circ,\ddagger}}{\partial p}\right)_T$$

- From the equation $(\partial G/\partial p)_T = V$, we can write

$$\left(\frac{\partial \Delta_r G^{\circ,\ddagger}}{\partial p}\right)_T = \Delta_r V^{\circ,\ddagger}$$

as the relationship holds equally for Δ_r quantities. $\Delta_r V^{\circ,\ddagger}$ is called the standard molar volume of activation and we interpret it as the volume change in going from reactants to the transition state. Then combining the two equations above to give

$$\left(\frac{\partial \ln k_{2nd}}{\partial p}\right)_T = \frac{-\Delta_r V^{\circ,\ddagger}}{RT}.$$

The effect of pressure: volumes of activation

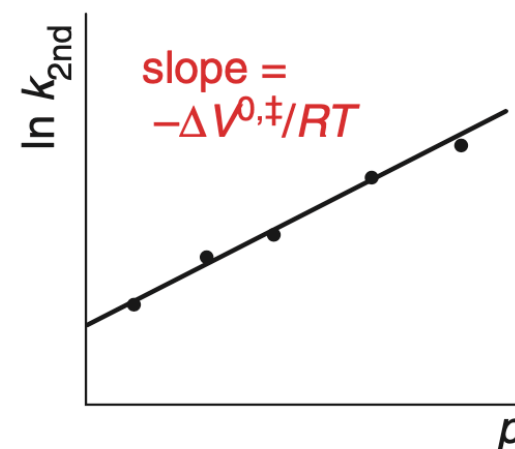
- We can integrate this using the assumption that $\Delta_r V^{\circ,\ddagger}$ *does not vary with pressure over the range of interest*:

$$\int d \ln k_{2\text{nd}} = \frac{-\Delta_r V^{\circ,\ddagger}}{RT} \int dp \quad (\text{const. temp.})$$
$$\ln k_{2\text{nd}} = \frac{-\Delta_r V^{\circ,\ddagger}}{RT} p + \text{const.}$$

- This implies that a plot of $\ln k_{2\text{nd}}$ against p will give a straight line of slope $-\Delta_r V^{\circ,\ddagger}/RT$. *Care is needed with the units here*: to work in SI p must be in N m^{-2} , which will yield $\Delta_r V^{\circ,\ddagger}$ in $\text{m}^3 \text{mol}^{-1}$; a common error is to leave the pressure in atmospheres.
- The integral can also be evaluated between limits, typically between p° (the standard pressure) and p to give

$$\ln \left(\frac{k_{2\text{nd}}(p)}{k_{2\text{nd}}(p^\circ)} \right) = \frac{-\Delta_r V^{\circ,\ddagger}}{RT} (p - p^\circ).$$

- Typical values of $\Delta_r V^{\circ,\ddagger}$ are not more than $10^{-5} \text{m}^3 \text{mol}^{-1}$ ($10 \text{cm}^3 \text{mol}^{-1}$); positive and negative values are found.



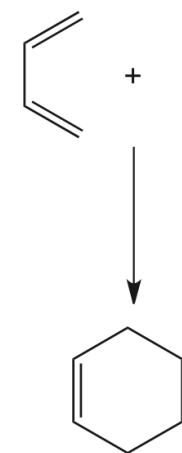
Typical plot used to determine the volume of activation.

Interpretation of volumes and entropies of activation

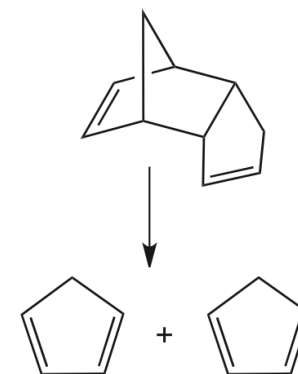
- Experimentally determined values of the volume and entropy of activation are used to find out about the mechanisms of reactions, especially those in solution. *It should be emphasised that in a complex reaction it is generally the rate determining step which determines the apparent entropy or volume of activation.*
- Since the enthalpy of activation is a close parallel to the activation energy, its value is interpreted in the same way. For example, a factor which is expected to lower the energy of the transition state will similarly lower the enthalpy of activation. *The entropy of activation is a new parameter for us and, as we shall see, its value gives some useful clues as to the form of the transition state.*

Gas-phase reactions

- For a bimolecular reaction two molecules go to one when the transition state is formed. *There is therefore a loss of translational degrees of freedom and so the process is expected to be accompanied by a significant decrease in the entropy.* The bimolecular (gas-phase) Diels–Alder reaction between butadiene and ethene is found to have an entropy of activation of $-150 \text{ J K}^{-1} \text{ mol}^{-1}$, in line with these expectations.
- *For unimolecular reactions there is no change in the number of molecules as the transition state is formed, so we do not expect there to be a large change in the entropy.* For example the unimolecular reverse Diels–Alder reaction shown opposite is found to have $\Delta_r S^{\circ, \ddagger} = -8 \text{ J K}^{-1} \text{ mol}^{-1}$.



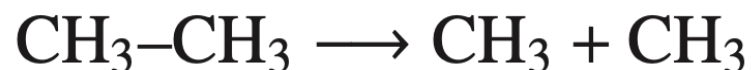
The Diels–Alder reaction



A reverse Diels–Alder reaction

Gas-phase reactions

- The first-order dissociation of ethane to methyl radicals



is found to have $\Delta_r \mathbf{S}^{\circ,\ddagger} = +58 \text{ J K}^{-1} \text{ mol}^{-1}$ in the gas phase.

This positive value of the entropy change means that the entropy of the transition state is higher than that of the reactant, even though both the transition state and the reactant are single molecules. The argument put forward here is that in the transition state the C–C bond is stretched and weakened when compared to the reactant. The transition state therefore has more available energy levels due to its internal motions, and hence has a higher entropy.

Gas-phase reactions

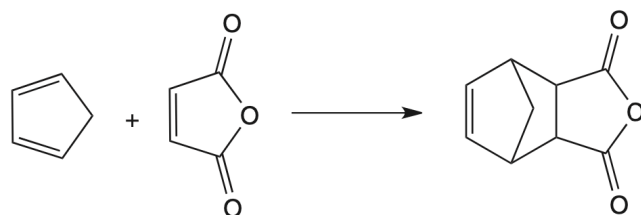
- In contrast, the first-order isomerization reaction



is found to have $\Delta_r S^{\circ,\ddagger} = -32 \text{ J K}^{-1} \text{ mol}^{-1}$ in the gas phase. *This negative value for the activation entropy means that the transition state has lower entropy than the reactant*, and the explanation for this is that this reaction is thought to proceed via a cyclic transition state in which the main atoms form a six-membered ring. The transition state is therefore more ‘constrained’ than the open-chain reactant, and so has less available energy levels and hence lower entropy.

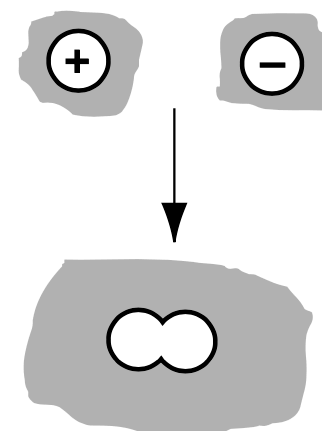
Reactions in solution

- The same issues are relevant for reactions in solution as for those in the gas phase. For example, the Diels–Alder reaction



has $\Delta_r S^{\circ,\ddagger} = -150 \text{ J K}^{-1} \text{ mol}^{-1}$ in dioxan solution – a value very comparable to that for similar reactions in the gas phase.

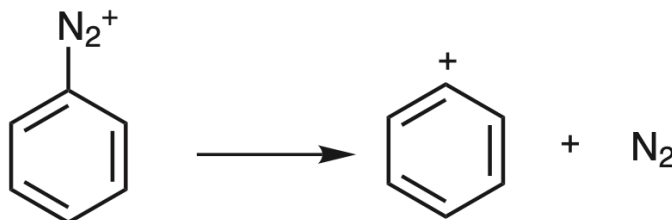
- In polar solvents reactions involving ionic species are common, and for such reactions we need to take account of the effects of solvation of the ions;* under some circumstances, these effects can be dominant. Consider a reaction in which two ions of opposite charge come together to form the transition state. The transition state is overall neutral so that the solvent molecules which were strongly co-ordinated by the ions are released. This release makes a positive contribution to $\Delta_r S^{\circ,\ddagger}$ which can be dominant over other factors. *The reduction in entropy of the solvent when it is co-ordinated to an ion is called electrostriction.*



When two oppositely charged ions come together to form a neutral transition state the co-ordinated solvent (shown schematically in grey) is no longer so tightly held. As a result the entropy of activation can be positive (as can the volume of activation).

Reactions in solution

- A more subtle effect is when formation of the transition state involves a spreading out of the charge. For example, the unimolecular decay (in water) of the benzene diazonium cation:



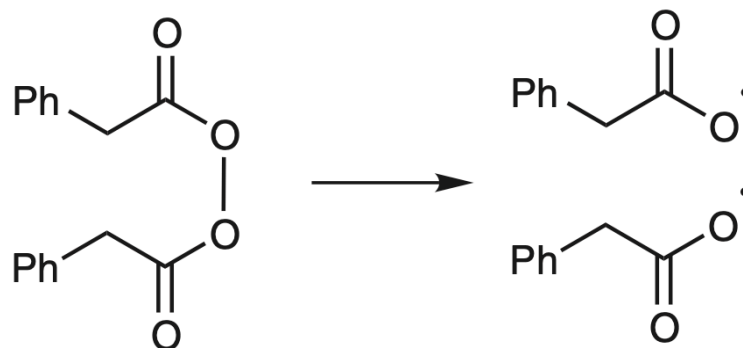
Reaction R1

which has $\Delta_r S^{\circ,\ddagger} = +44 \text{ J K}^{-1} \text{ mol}^{-1}$.

- The interpretation here is that in the transition state the positive charge moves from being located on the N to being spread between the C and the N. There is thus a *dispersal* of charge leading to less electrostriction of the solvent and so an increase in entropy. In fact, this observation of a positive $\Delta_r S^{\circ,\ddagger}$ is taken as evidence that the reaction proceeds by this unimolecular decay. The alternative process, involving direct attack by H_2O on the cation is associative and certainly not consistent with a positive $\Delta_r S^{\circ,\ddagger}$.

Interpretation of the volume change

- A unimolecular reaction is expected to have a positive volume of activation simply as the transition state is 'larger' on account of some of the bonds beginning to stretch prior to dissociation. For example, the dissociation of benzoyl peroxide (in CCl_4):

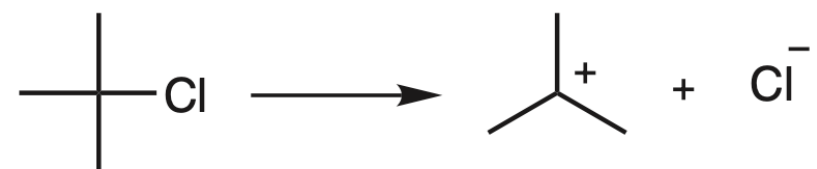


has $\Delta_r V^{\circ,\ddagger} = +10 \text{ cm}^3 \text{ mol}^{-1}$.

- As with the entropy change, electrostriction effects can be important in determining the volume of activation of reactions involving ions.* For example, a reaction involving charge dispersal is expected to have a positive volume of activation e.g. reaction [R1] in previous slide has $\Delta_r V^{\circ,\ddagger} = +10 \text{ cm}^3 \text{ mol}^{-1}$.

Interpretation of the volume change

- In contrast, a reaction in which there is charge *separation* in the transition state leads to more electrostriction and hence a negative volume of activation. For example, the formation of a carbocation (in EtOH/H₂O solution):



has $\Delta_r V^{\circ,\ddagger} = -16 \text{ cm}^3 \text{ mol}^{-1}$. We can attribute this to the build up of charge on the Cl and the C in the transition state as the C–Cl bond prepares to break heterolytically. These volumes of activation have been used to differentiate between a dissociative mechanism (like S_N1) and an associative one (like S_N2).

- Since much the same factors are involved in determining the size and sign of the entropy and volume of activation, it is not surprising to find that $\Delta_r V^{\circ,\ddagger}$ and $\Delta_r S^{\circ,\ddagger}$ values are well correlated.

Substitution reactions in transition metal complexes

- Reactions in which one ligand X in a transition metal complex is substituted by another species Y:



provide good examples of how values of the entropy and volume of activation can be used to probe reaction mechanisms.

- For these reactions we can imagine three basic mechanisms. In the *dissociative* mechanism X leaves the complex in one step (usually this is the rate-determining step, the RDS), and then Y attaches itself in a subsequent step



- In the *associative* mechanism Y attaches itself to the complex in the first step (usually the RDS), and then X leaves in a subsequent step



- In the *interchange* mechanism the arrival of Y and the departure of X are concerted

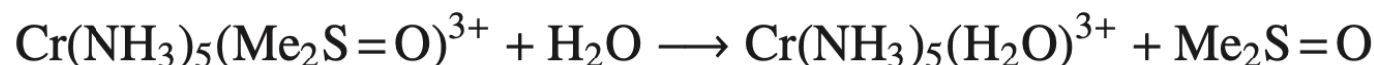


Substitution reactions in transition metal complexes

- So if we ignore effects due to charges, then we would expect the following
 - **dissociative mechanism:** as a molecule splits apart in the RDS, we expect both the volume and entropy of activation to be *positive*.
 - **associative mechanism:** in the RDS two molecules come together to form the transition state; we therefore expect that the volume and entropy of activation will be *negative*.
 - **interchange mechanism:** two molecules come together to form the transition state, but there is a 'loosening' of this transition state as X starts to leave. Typically the entropy of activation is close to zero, but the volume of activation could have either sign.

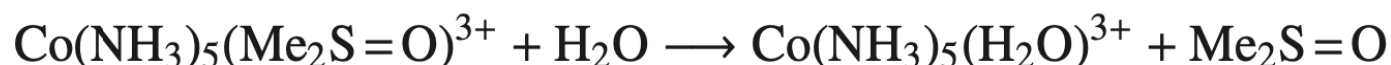
Substitution reactions in transition metal complexes

- The substitution reaction of the Cr(III) complex



is found to have $\Delta_r S^{\circ,\ddagger} = -15 \text{ J K}^{-1} \text{ mol}^{-1}$ and $\Delta_r V^{\circ,\ddagger} = -3.2 \text{ cm}^3 \text{ mol}^{-1}$. This is consistent with the reaction proceeding via the associative mechanism.

- In contrast, the similar substitution reaction of the Co(III) complex



is found to have $\Delta_r S^{\circ,\ddagger} = +10 \text{ J K}^{-1} \text{ mol}^{-1}$ and $\Delta_r V^{\circ,\ddagger} = +2.0 \text{ cm}^3 \text{ mol}^{-1}$. This is consistent with the reaction proceeding via the dissociative mechanism. Substitution reactions at square planar complexes (e.g. PtX_4) are found to have negative values of $\Delta_r S^{\circ,\ddagger}$, which point to an associative mechanism.

- *Generally speaking it is thought that the values of the volume of activation are better indicators of mechanism than the values of the entropy of activation.*

Course outline

- Boltzmann distribution

Boltzmann distribution

- In the introduction we mentioned the Boltzmann distribution, which predicts the number of particles, n_i , in a level with energy ϵ_i as

$$n_i = \frac{N}{q} \exp\left(\frac{-\epsilon_i}{kT}\right)$$

where N is the total number of particles and q is the molecular partition function. We can now derive this distribution law from the canonical distribution function in the following way; *the argument applies to non-interacting particles.*

Boltzmann distribution

- Recall that the internal energy can be calculated from the molecular partition function:

$$U = NkT^2 \left(\frac{\partial \ln q}{\partial T} \right)_V = NkT^2 \frac{1}{q} \left(\frac{\partial q}{\partial T} \right)_V$$

- The derivative of q can be computed term by term:

$$\left(\frac{\partial q}{\partial T} \right)_V = \frac{\partial}{\partial T} \left(\sum_i \exp \left(\frac{-\epsilon_i}{kT} \right) \right)_V = \frac{1}{kT^2} \sum_i \epsilon_i \exp \left(\frac{-\epsilon_i}{kT} \right)$$

where we have used the fact that the energy levels are constant at fixed volume.

- Substituting this into the expression for U gives

$$U = NkT^2 \frac{1}{q} \left\{ \frac{1}{kT^2} \sum_i \epsilon_i \exp \left(\frac{-\epsilon_i}{kT} \right) \right\} = \frac{N}{q} \sum_i \epsilon_i \exp \left(\frac{-\epsilon_i}{kT} \right).$$

- An alternative way of calculating the internal energy, U , would be to take the population of each state, multiply this by the energy of that state and then sum over all states

$$U = \sum_i n_i \epsilon_i.$$

Boltzmann distribution

- Writing out a few terms from each of equations for **U** we find

$$U = \frac{N}{q} \varepsilon_0 \exp\left(\frac{-\varepsilon_0}{kT}\right) + \frac{N}{q} \varepsilon_1 \exp\left(\frac{-\varepsilon_1}{kT}\right) + \frac{N}{q} \varepsilon_2 \exp\left(\frac{-\varepsilon_2}{kT}\right) + \dots$$

$$U = n_0 \varepsilon_0 + n_1 \varepsilon_1 + n_2 \varepsilon_2 + \dots$$

- Equating each term in the first of these expressions by the corresponding one in the second we find

$$n_i = \frac{N}{q} \exp\left(\frac{-\varepsilon_i}{kT}\right)$$

- *which is the Boltzmann distribution. This applies equally well to distinguishable and indistinguishable particles*; it was first derived as a result in classical physics (in which the energy is continuous) and is sometimes known as *the Maxwell–Boltzmann distribution*.

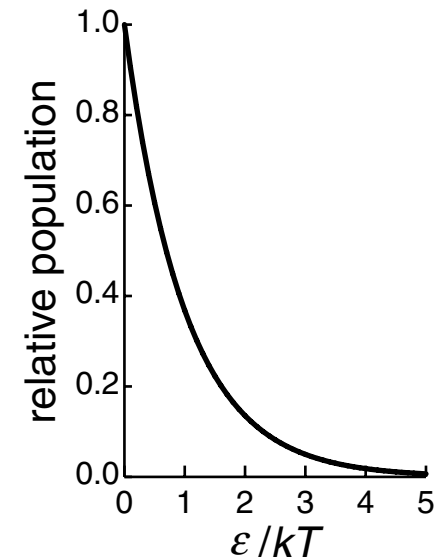
Boltzmann distribution

- As was commented before, what the distribution says is that only those states whose energy is less than kT are populated significantly. The lowest energy state is always the most populated, and the population falls off as the energy increases.
- Sometimes it is convenient to consider levels rather than states, in which case it is necessary to include a degeneracy term, g_j

$$n_j = \frac{Ng_j}{q} \exp\left(\frac{-\varepsilon_j}{kT}\right).$$

- The Boltzmann distribution law has numerous applications in physical science, especially in spectroscopy.

$$n_i = \frac{N}{q} \exp\left(\frac{-\varepsilon_i}{kT}\right)$$



The Boltzmann distribution predicts that the relative population of a level (n_i/n_0) falls off as its energy increases. The population depends on the dimensionless ratio ε/kT .

Applications of the Boltzmann distribution

- Interpretation of the partition function
- Populations of vibrational energy levels
- Populations of rotational energy levels
- Non-equilibrium states

Interpretation of the partition function

- We can use the Boltzmann distribution to find a useful interpretation of the partition function. Suppose that the ground state has energy zero, $\epsilon_0 = 0$, then from the equation

$$n_0 = \frac{N}{q}$$

and so

$$q = \frac{N}{n_0}$$

- If all of the particles are in the ground state, $n_0 = N$ and $q = 1$. As particles leave the ground state and move up into higher energy levels n_0 decreases and so q increases. The partition function can therefore be interpreted as a measure of the fraction of particles which have left the ground state.

Populations of vibrational energy levels

- If we use a simple harmonic oscillator as the model, the energy levels are

$$E_v = v h \nu_0$$

where ν_0 is the vibration frequency (in Hz) and the lowest level, $v = 0$, has been taken as the energy zero.

- The vibrational partition function is given by the equation

$$q'_{\text{vib}} = \frac{1}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)}$$

where $\theta_{\text{vib}} = h\nu_0/k$.

- Using these in the equation, $n_0 = N/q$, the population of the vibrational energy levels is

$$\begin{aligned} n_v &= N \left[1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right) \right] \exp\left(\frac{-v h \nu_0}{kT}\right) \\ &= N \left[1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right) \right] \exp\left(\frac{-v \theta_{\text{vib}}}{T}\right) \end{aligned}$$

Example

- Calculate the populations of the first three vibrational levels of I_2 at 298 K, which has a vibrational frequency of 214 cm^{-1} .

Populations of rotational energy levels

- If we use the rigid rotor as a model for the energy levels of a diatomic molecule, the energies are

$$E_J = BJ(J + 1)$$

where B is the rotational constant.

- These levels have a degeneracy of **(2 J + 1)**, so the appropriate form of the Boltzmann distribution is

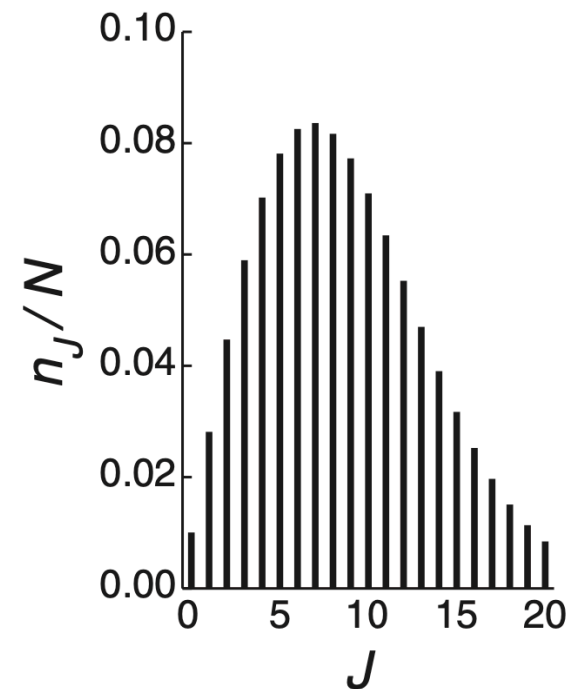
$$n_J = \frac{N(2J + 1)}{q_{\text{rot}}} \exp\left(\frac{-BJ(J + 1)}{kT}\right)$$

- If we assume that **B** is much smaller than **kT** (which at room temperature is almost always the case), then **$q_{\text{rot}} = T/\sigma\theta_{\text{rot}}$** , where **$\theta_{\text{rot}} = B/k$** . Then the distribution becomes

$$\frac{n_J}{N} = \frac{\sigma\theta_{\text{rot}}}{T} (2J + 1) \exp\left(\frac{-\theta_{\text{rot}}J(J + 1)}{T}\right)$$

Populations of rotational energy levels

- As J increases (i.e. as we go to higher energy levels) there are two terms which affect the population in different ways. The degeneracy term, $(2J + 1)$, increases with increasing J , but the exponential term decreases with increasing J . At first the degeneracy term wins and the populations rise with increasing J . Eventually, however, the exponential dominates and the populations fall off – the result is a maximum. The bar chart opposite illustrates this for CO at 2 98 K ($\tilde{B} = 1.925 \text{ cm}^{-1}$, $\theta_{\text{rot}} = 2.77 \text{ K}$); note the rise to a maximum. The form of this population distribution explains the intensity distribution in the P and R branches seen in the IR spectrum of diatomic molecules.



Populations of rotational energy levels

- We can find the value of J at which the population is a maximum by differentiating the expression for n_J with respect to J (and in the process supposing that J is a continuous variable). The maximum is found by setting this derivative equal to zero

$$\begin{aligned}\frac{d}{dJ}n_J &= \frac{d}{dJ} \left[N \frac{\sigma \theta_{\text{rot}}}{T} (2J + 1) \exp \left(\frac{-\theta_{\text{rot}} J (J + 1)}{T} \right) \right] \\ &= 0 \quad \text{for a maximum.}\end{aligned}$$

- The maximum is found to be given by

$$J_{\text{max}} = \sqrt{\frac{T}{2\theta_{\text{rot}}}} - \frac{1}{2}$$

- For the case of CO at 298 K, J is 6.8; to find the integer value with the largest population it is necessary to return to the expression for n_J . In this case it is found that the maximum is at $J = 7$.

Non-equilibrium states

- *The Boltzmann distribution predicts the populations of the energy levels at equilibrium.* These populations can be altered from their equilibrium values by such things as chemical reactions or intense radiation (particularly from a laser). Once the perturbation is removed, we expect that equilibrium, and hence the Boltzmann distribution, will eventually be restored.
- Given a particular set of energy levels, the way in which the particles distribute themselves amongst the levels depends only on the temperature. So, we can say that *the distribution is characteristic of the temperature.* To find the temperature we can plot $\ln n_i$ against ϵ_i ; according to the Boltzmann distribution this will have a slope of $-1/kT$.

Non-equilibrium states

- A two-level system is a system with only two energy levels; there are a number of molecular systems which can be modelled as two-level systems over a limited temperature range.
- In a two-level system the ratio of the populations in the upper and lower levels, n_1 and n_0 , is, according to the Boltzmann distribution,

$$\frac{n_1}{n_0} = \exp\left(\frac{-\Delta\varepsilon}{kT}\right)$$

where $\Delta\varepsilon$ is the energy separation of the two levels.

- At thermal equilibrium $n_0 > n_1$ as $\Delta\varepsilon$ and T are both positive. However, if we make $n_1 > n_0$ (for example by applying intense radiation which causes transitions between the two levels), the above equation predicts that the temperature is negative. This situation is called a population inversion.

Density of states

- As we have already seen, the translational energy levels are very closely spaced compared to kT . A useful way of describing this is to compute a quantity known as the *density of states*.
- *The density of states, $D(\epsilon)$, gives the number of energy levels present at energy ϵ ; to be precise $D(\epsilon)d\epsilon$ is the number of states in the interval ϵ to $\epsilon + d\epsilon$.* It is calculated by first finding the number of states, $W(\epsilon)$, which have energy less than or equal to ϵ . The number of states between ϵ and $\epsilon + d\epsilon$ is $[W(\epsilon + d\epsilon) - W(\epsilon)]$. Hence

$$\begin{aligned} D(\epsilon)d\epsilon &= W(\epsilon + d\epsilon) - W(\epsilon) \\ \text{hence } D(\epsilon) &= \frac{W(\epsilon + d\epsilon) - W(\epsilon)}{d\epsilon} \\ &= \frac{dW(\epsilon)}{d\epsilon} \end{aligned}$$

- The first task is then to find an expression for $W(\epsilon)$, the number of states with energy $\leq \epsilon$.

Density of states

- We can model the translational energy levels of a molecule by using those of a particle in a three-dimensional box. If we assume that the box is cubical and of side a the energy levels are

$$E_{n_x, n_y, n_z} = (n_x^2 + n_y^2 + n_z^2) \frac{h^2}{8ma^2}$$

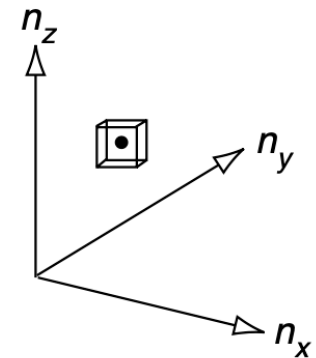
where the three quantum numbers n_x , n_y and n_z can take values 1, 2, 3

- In the one-dimensional case it is easy to find the value of n_x which corresponds to a certain energy and hence find the number of states. However, in the three-dimensional case counting the energy levels is not as straight forward as the system shows degeneracy i.e. there is more than one state with the same energy. For example the state with $(n_x, n_y, n_z) = (1, 2, 3)$ is degenerate with the states $(2, 1, 3)$, $(3, 1, 2)$ and $(3, 2, 1)$. Counting all of these states needs a clever 'trick'.

Density of states

- The way to proceed is to note that each energy level (state) has a unique set of values of these three quantum numbers, and can be thought of as being represented by a point in a three dimensional space whose axes give the values of n_x , n_y and n_z (illustrated opposite). We can think of the point as lying at the centre of a cube with unit dimensions along each axis.
- The energy of the state represented by this point depends on $(n_x^2 + n_y^2 + n_z^2)$ which can be thought of as the square of the distance of the point from the origin; let this distance squared be written n^2 . The energy can then be written as

$$E_n = \frac{n^2 h^2}{8ma^2}$$



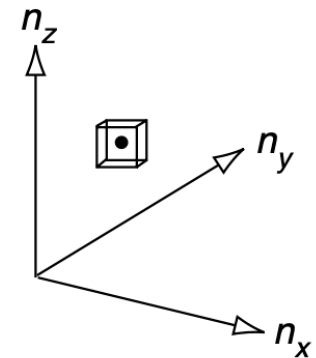
Density of states

- In our diagram, all of the states with energy $\leq \varepsilon$ are represented by points which lie within a distance $n_{\max}$ of the origin where

$$n_{\max} = \left(\frac{8ma^2\varepsilon}{h^2} \right)^{\frac{1}{2}}$$

$$\varepsilon = \frac{n_{\max}^2 h^2}{8ma^2}$$

- The number of states within this distance $n_{\max}$ is one eighth of the volume of a sphere with radius $n_{\max}$. The reason for this is that as each state occupies a unit volume, the number of states is equal to the volume. However, only one eighth of the sphere must be included as we know that the quantum numbers n_x , n_y and n_z are *positive integers*. So only that part of the sphere in which all these quantum numbers are positive is needed – this is one eighth of the total.



Density of states

- Hence

$$\begin{aligned}\text{no. of states} &= \frac{1}{8} \text{ vol. of sphere} \\ &= \frac{1}{8} \frac{4}{3} \pi n_{\text{max}}^3 \\ &= \frac{1}{8} \frac{4}{3} \pi \left(\frac{8ma^2 \varepsilon}{h^2} \right)^{\frac{3}{2}}.\end{aligned}$$

- This expression can be tidied up by writing the volume, V , (of the box) as a^3 to give

$$W(\varepsilon) = \frac{\pi}{6} \left(\frac{8m\varepsilon}{h^2} \right)^{\frac{3}{2}} V.$$

- The density of states is found by differentiating this with respect to ε

$$\begin{aligned}D(\varepsilon) &= \frac{dW(\varepsilon)}{d\varepsilon} = \frac{d}{d\varepsilon} \left[\frac{\pi}{6} \left(\frac{8m\varepsilon}{h^2} \right)^{\frac{3}{2}} V \right] \\ &= \frac{3}{2} \frac{\pi}{6} \left(\frac{8m}{h^2} \right)^{\frac{3}{2}} \varepsilon^{\frac{1}{2}} V \\ &= \frac{\pi}{4} \left(\frac{8m}{h^2} \right)^{\frac{3}{2}} \varepsilon^{\frac{1}{2}} V.\end{aligned}$$

Example

- For a sample of He gas at 298 K and at pressure 1 bar in a container of volume 1 dm³, calculate the number of states with energy $\leq kT$. Compare this with the number of atoms.

Example

- For the He sample, calculate the density of states at energy kT in a volume of 1 dm^3 and at 298K

Summary of last lecture

- Eyring equation from transition state theory
 - $k_{2nd} = (1/c^\circ) (kT/h) K^\ddagger$
 - Link to thermodynamics via $\Delta_r G^\ddagger = -RT \ln K^\ddagger$ and $\Delta_r G^\ddagger = \Delta_r H^\ddagger - T \Delta_r S^\ddagger$
 - Eyring plot: $\ln (k_{2nd} / T)$ against $1/T$
- Arrhenius equation
 - Activation energy defined by local slope of $\ln k_{2nd}$ versus $1/T$
 - Relation of $\Delta_r H^\ddagger$ to E_a depends on conditions.
- Volumes and entropies of activation
 - Relate to formation of the transition state from the reactants
 - Loss of degrees of freedom (e.g. associative mechanisms) gives $\Delta_r S^\ddagger < 0$
 - Electrostriction: coordination of solvent around a charge decreases entropy.

Summary of last lecture

- Interpretation of the Boltzmann distribution
 - A plot of $\ln n_i$ against ϵ_i has slope $-1/kT$ at equilibrium, thereby defining temperature.
 - For a population inversion, the effective temperature is negative.
- Density of states $D(\epsilon)$
 - The number of states lying within a small energy interval is $D(\epsilon) dE$.
 - Related to the total number of states up to ϵ , $W(\epsilon)$ by $D(\epsilon) = dW(\epsilon)/d\epsilon$.
 - Density of states for 3D particle-in-a-box:
 - “State space” for the three quantum numbers n_x, n_y, n_z .
 - States with the same energy lie on a sphere in this state space.
 - $W(\epsilon)$ is the volume of the positive octant of a sphere in state space.